

Hostile Review: Minimax Lower Bound Proof (Theorem 4)

Document reviewed: minimax_lower_bound_proof.md **Reviewer stance:** Annals of Statistics hostile reviewer **Overall verdict:** The central claim (rate optimality, exponent $2M\Delta^2$) is **likely correct in spirit but rests on an invalid inequality in its critical step**. The proof as written is **not acceptable** for publication.

Summary of Fatal and Non-Fatal Issues

Issue	Severity	Status
Lemma 7 (Slud’s inequality) is false	Fatal	The claimed inequality is not a consequence of Slud (1977) and fails numerically for many parameter values
K=2 to K>2 reduction conflates bound truth with bound looseness	Major	The argument that “K=2 is hardest” uses direction of an inequality opposite to what is needed

HOSTILE REVIEW: MINIMAX LOWER BOUND PROOF (THEOREM 4)2

Issue	Severity	Status
F1 conversion (Lemma 8) derivation is algebraically unsound	Major	The claimed inequality $1 - \text{F1} \geq 1 - \text{TPR}$ has an unverified validity condition
Mixture-product TV bound (Lemma 4)	Correct	Convexity argument is valid; direction is appropriate for Le Cam
Le Cam construction and symmetry	Correct	Construction is within the allowed class under A6 with $C_{\text{bal}} = 1$
Cramer-Chernoff large deviations	Correct	Standard asymptotic analysis
χ^2 tensorization	Correct	Convexity bound handles mixture before tensorization; no independence violation
η factor gap	Incomplete	Acknowledged by author, but leaves the η characterization unresolved

1. Lemma 7 (Slud's Inequality) is False as Stated

Claim (Lemma 7). Let $S_M \sim \text{Binomial}(M, p)$ with $p \leq 1/2$. For any k with $p \leq k/M \leq 1/2$:

$$\mathbb{P}(S_M \geq k) \geq \frac{1}{2} \exp\left(-2M\left(\frac{k}{M} - p\right)^2\right).$$

This inequality is not a valid consequence of Slud (1977) and is numerically false.

1.1 What Slud Actually Proves

Slud (1977), Theorem 2, gives a **Gaussian comparison** inequality:

$$\mathbb{P}(S_M \geq k) \geq \frac{1}{2} \Phi\left(-\frac{k - Mp}{\sqrt{Mp(1-p)}}\right),$$

where Φ is the standard normal CDF. This is a genuine result. However, the further step of bounding the Gaussian tail by $(1/2) \exp(-2M\delta^2)$ uses the inequality

$$\Phi(-t) \geq \frac{1}{\sqrt{2\pi}} \frac{t}{1+t^2} e^{-t^2/2}, \quad t > 0.$$

Combining these gives

$$\mathbb{P}(S_M \geq k) \geq \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} \frac{t}{1+t^2} \exp(-t^2/2)$$

where $t = \delta\sqrt{M}/\sqrt{p(1-p)}$ and $\delta = k/M - p$. The exponent here is $t^2/2 = M\delta^2/(2p(1-p))$, **not** $2M\delta^2$. The two exponents differ by the factor $1/(4p(1-p))$.

Since $p(1-p) \leq 1/4$, we have $1/(4p(1-p)) \geq 1$, meaning

$$\exp(-M\delta^2/(2p(1-p))) \leq \exp(-2M\delta^2).$$

The **inequality goes the wrong direction**: Slud's actual bound gives a Gaussian exponent that is *larger* than $2M\delta^2$, producing a *smaller* RHS, not a larger one. The author attempts to replace the correct Gaussian exponent $M\delta^2/(2p(1-p))$ with the smaller Hoeffding exponent $2M\delta^2$ while *simultaneously* claiming a *larger* lower bound. This is algebraically impossible without a compensating factor that would be enormous (roughly $\exp(M\delta^2 \cdot (1/(2p(1-p)) - 2))$), and no such factor appears.

1.2 Numerical Counterexamples

The claimed inequality fails for multiple parameter settings:

Example 1: $M = 20$, $p = 0.4$, $k = 10$ (so $\delta = 0.1$).

$$\mathbb{P}(\text{Bin}(20, 0.4) \geq 10) \approx 0.244, \quad \text{RHS} = 0.5 \cdot e^{-40 \cdot 0.01} = 0.5 \cdot e^{-0.4} \approx 0.335.$$

$0.244 \not\geq 0.335$. The inequality is violated by nearly 40%.

Example 2: $M = 10$, $p = 0.3$, $k = 5$ ($\delta = 0.2$).

$$\mathbb{P}(\text{Bin}(10, 0.3) \geq 5) \approx 0.151, \quad \text{RHS} = 0.5 \cdot e^{-20 \cdot 0.04} = 0.5 \cdot e^{-0.8} \approx 0.225.$$

$0.151 \not\geq 0.225$.

Example 3: $M = 100$, $p = 0.4$, $k = 50$ ($\delta = 0.1$).

$$\mathbb{P}(\text{Bin}(100, 0.4) \geq 50) \approx 0.027, \quad \text{RHS} = 0.5 \cdot e^{-200 \cdot 0.01} = 0.5 \cdot e^{-2} \approx 0.068.$$

$0.027 \not\geq 0.068$.

These are not edge cases; they are squarely in the parameter regime the proof requires ($p = \mu < 1/2$, $\delta = 1/2 - \mu$).

1.3 Ceiling Issue for Odd M

The proof applies Slud's inequality with $k = \lceil M/2 \rceil$, but Slud's condition requires $k/M \leq 1/2$. For odd M , $\lceil M/2 \rceil/M = (M+1)/(2M) = 1/2 + 1/(2M) > 1/2$, violating the premise. The author silently replaces $\lceil M/2 \rceil/M$ with $1/2$ in the exponent, which is an approximation that does not respect the inequality's domain.

1.4 Impact on the Main Proof

This is not a minor technical issue. The Binomial tail lower bound is **the entire content** of the testing lower bound for $K = 2$. If Lemma 7 is false, the claimed exponent $2M\Delta^2$ is unsubstantiated. The correct Slud-based bound gives

$$\mathbb{P}(\text{Bin}(M, \mu) \geq M/2) \geq \frac{1}{2\sqrt{2\pi}} \frac{t}{1+t^2} \exp\left(-\frac{M(1/2-\mu)^2}{2\mu(1-\mu)}\right),$$

which has exponent $M\Delta^2/(2\mu(1-\mu))$ rather than $2M\Delta^2$. Since $\mu(1-\mu) \leq 1/4$, the correct exponent is *at least* $2M\Delta^2$ (i.e., the same or faster decay), but the prefactor depends on μ and M and the bound is much weaker in finite samples.

The author needs to prove a lower bound, and a bound that decays *faster* (with larger exponent) is a *stronger* lower bound for that exponent. Actually, wait – a larger exponent means the RHS is *smaller*, which gives a *weaker* lower bound on the probability. So the correct Slud bound gives a *smaller* RHS than what the author claims. This makes the author’s bound an *overestimate* of the true minimal possible RHS.

To salvage the result, the author would need to show that the minimax risk is at least $(1/2)\exp(-2M\Delta^2)$ even though $\mathbb{P}(\text{Bin}(M, \mu) \geq M/2)$ may be smaller than this. This would require a different argument entirely, as the current proof relies on the optimal Bayes test having error exactly $\mathbb{P}(\text{Bin}(M, \mu) \geq M/2)$.

1.5 Recommendation

Lemma 7 must be either (a) replaced with the correct Slud inequality and the Gaussian tail bound, which would yield a different (potentially μ -dependent) exponent; or (b) proved from first principles if the claimed form is actually true (which the counterexamples above suggest it is not). The author should check whether Slud’s proof can be adapted to yield the claimed bound for the specific case $p = \mu$, $k = M/2$ – perhaps via a sharper inequality that does not go through the Gaussian approximation – but as it stands, the proof does not provide this.

2. The $K=2$ to $K>2$ Reduction is Logically Inverted

Claim (Section 4.6). For $K > 2$, the per-component χ^2 divergence is larger than for $K = 2$, and “since larger χ^2 makes the distributions more distinguishable, the $K = 2$ case is the hardest.”

2.1 The Argument’s Structure

The author writes:

For $K > 2$: $\chi_{\text{bern}}^2 = \frac{4(\Delta^*)^2}{(1-\mu/(K-1)) \cdot \mu/(K-1)} \geq 16(\Delta^*)^2$ This is *larger* than the $K = 2$ case. Since larger χ^2 makes the distributions more distinguishable, the $K = 2$ case is the hardest.

Then concludes the $K = 2$ lower bound holds for $K > 2$.

2.2 The Gap

For $K = 2$, the noise distribution P_1 is a **product** Bernoulli $(1 - \mu)^{\otimes M}$. For $K > 2$, P_1 is a **mixture** $\frac{1}{K-1} \sum_{c \neq y^*} \text{Bernoulli}(1 - \mu/(K-1))^{\otimes M}$. The author uses Lemma 4 to bound

$$\text{TV}(P_0, P_1) \leq \sqrt{\frac{(1 + \chi_{\text{bern}}^2)^M - 1}{2}}.$$

This is an **upper bound** on TV. For the purpose of the Le Cam lower bound $\inf R \geq (1 - \text{TV})/2$, a *smaller* upper bound on TV gives a *stronger* lower bound on risk. A *larger* upper bound (from larger χ^2) gives a *weaker* lower bound.

The author’s argument treats the χ^2 value as if it directly measures the difficulty of distinguishing P_0 from P_1 . But the bound $\text{TV} \leq \sqrt{((1 + \chi^2)^M - 1)/2}$ is not tight in general; it becomes looser as χ^2 grows. The author confuses:

- **The bound on TV** (which becomes larger, i.e., less informative, for $K > 2$), with
- **The actual TV** (which may be smaller or larger for $K > 2$; the bound doesn’t tell us).

For a concrete illustration: if $\chi_1^2 < \chi_2^2$, it is entirely possible that $\sqrt{((1 + \chi_2^2)^M - 1)/2} \gg 1$ (bound useless) while the actual $\text{TV}(P_0, P_1)$ is close to 0 (very hard to distinguish). The bound does not rule this out.

2.3 Why the Conclusion is Probably Correct (But Not Proved Here)

The $K = 2$ case is likely the hardest because under $C_{\text{bal}} = 1$, all mixture components for $K > 2$ are identical (same parameter $1 - \mu/(K - 1)$), making P_1 a single product distribution, not a true mixture. For $K > 2$, the “gap” between μ and $1 - \mu/(K - 1)$ is larger, making the product distributions more separated. This intuition is correct, but the proof as written **does not make this argument**. Instead, it invokes the χ^2 bound in a way that is logically reversed. The author should simply note that under $C_{\text{bal}} = 1$, P_1 is a product (all components equal), and the per-expert TV is $|1 - \mu K/(K - 1)|$, which exceeds $|1 - 2\mu|$ for $K > 2$, so $K = 2$ gives the smallest separation (hence hardest). No χ^2 bounding is needed.

3. The F1 Lower Bound Derivation is Algebraically Unsound

3.1 The Claim

Lemma 8 states:

$$1 - F1(\psi) \geq \eta \cdot \mathbb{P}(\psi = 0 \mid \text{noise}) + (1 - \eta) \cdot \mathbb{P}(\psi = 1 \mid \text{clean}),$$

but the derivation that follows does not establish this. Instead, the author ends with the weaker inequality

$$1 - F1(\psi) \geq \mathbb{P}(\psi = 0 \mid \text{noise}).$$

3.2 Tracing the Derivation

The author writes the F1 expression, then performs several algebraic manipulations that appear to be in **draft form** – incomplete cancellations, substitutions that don't match definitions, and a denominator identity that is incorrect. Specifically:

The author writes: “Denominator = $\eta + \eta \cdot \text{TPR} + (1 - \eta)\text{FNR}$ ”.

But the actual denominator of $1 - F1$ after substituting $\text{TP} = \eta \cdot \text{TPR}$, $\text{FP} = (1 - \eta)\text{FPR}$, $\text{FN} = \eta(1 - \text{TPR})$ is:

$$2\text{TP} + \text{FP} + \text{FN} = 2\eta \cdot \text{TPR} + (1 - \eta)\text{FPR} + \eta(1 - \text{TPR}) = \eta(1 + \text{TPR}) + (1 - \eta)\text{FPR}.$$

This does not match the author’s “ $\eta + \eta \cdot \text{TPR} + (1 - \eta)\text{FNR}$ ” unless $\text{FNR} = 1 - \text{TPR} = \text{FPR}$, which is not generally true. The derivation contains an algebraic inconsistency.

3.3 The Hidden Condition

The final inequality $1 - \text{F1} \geq 1 - \text{TPR}$ is equivalent (by cross-multiplication, as shown below) to

$$\frac{\text{FP} + \text{FN}}{2\text{TP} + \text{FP} + \text{FN}} \geq \frac{\text{FN}}{\text{TP} + \text{FN}} \iff \text{TP}(\text{FP} - \text{FN}) \geq 0 \iff \text{FP} \geq \text{FN}.$$

Substituting $\text{FP} = (1 - \eta)\text{FPR}$, $\text{FN} = \eta(1 - \text{TPR})$, the condition becomes

$$(1 - \eta)\text{FPR} \geq \eta(1 - \text{TPR}).$$

For the author’s construction, $\text{FPR} \approx \text{FNR} = 1 - \text{TPR}$ (they differ by at most $P(S = M/2)$), so the condition requires $\eta \leq 1/2$, which holds by assumption. But the author **never checks this condition**. The inequality is presented as if it holds universally, which it does not. For a detector with $\text{FPR} \ll \text{FNR}$ (e.g., a conservative test that rarely flags), the inequality can fail.

3.4 Why This Matters

The $1/\eta$ factor in the upper bound (Theorem 1) creates a potential gap: the upper bound degrades as $\eta \rightarrow 0$, but the lower bound’s η dependence is lost by the one-sided inequality $1 - \text{F1} \geq 1 - \text{TPR}$. A proper F1 lower bound

should retain the η factor to match the upper bound’s scaling. The author acknowledges this in a remark but does not resolve it.

4. Other Issues

4.1 Lemma 4 (Mixture-Product TV Bound) – Correct

The convexity inequality $\text{TV}(P_0, \frac{1}{L} \sum_{\ell} Q_{\ell}) \leq \frac{1}{L} \sum_{\ell} \text{TV}(P_0, Q_{\ell})$ is valid: TV is jointly convex, so it is convex in each argument. The bound direction (upper bound on TV) is correct for the Le Cam application. For the symmetric case ($C_{\text{bal}} = 1$), all Q_{ℓ} are equal, so the inequality is tight. No issue here.

4.2 Le Cam Construction – Correct

The construction is within the SCX assumption class. Choosing the hardest instance within $\mathcal{P}_{\Delta}$ is standard for minimax lower bounds. The symmetry assumption $C_{\text{bal}} = 1$ is restrictive but permissible for a lower bound. The construction correctly satisfies A1-A6.

4.3 Cramer-Chernoff Analysis – Correct

The asymptotic analysis is standard and correctly identifies $\text{KL}(1/2\|\mu)$ as the exact error exponent for the optimal Bayes test. The Taylor expansion $\text{KL}(1/2\|1/2-\Delta) = 2\Delta^2 + O(\Delta^4)$ is correct. The claim that $2\Delta^2$ is the worst-case (minimum) exponent over μ is also correct, since $\mu(1-\mu) \leq 1/4$ implies $\text{KL} \geq 2\Delta^2$. The concern raised in the review prompt about “the adversary

chooses the threshold” is not relevant here, as the author is analyzing the optimal Bayes test (which is likelihood-ratio based) rather than the threshold-based SCX detector.

4.4 χ^2 Tensorization – Correct

The tensorization $\chi^2(\prod P_m \| \prod Q_m) = \prod (1 + \chi^2(P_m \| Q_m)) - 1$ is well-known for products. The proof handles the mixture case by first applying the convexity bound (Lemma 4), which reduces the mixture to a sum of per-component TV terms. Each term involves a product Q_ℓ , so the tensorization applies to each term individually. There is no independence violation.

4.5 The η Factor Gap

The upper bound from Theorem 1 contains a $1/\eta$ factor: $F1 \geq 1 - (1/\eta) \exp(-2M\Delta^2)$. The lower bound in Theorem 4 gives $1 - F1 \geq (1/2) \exp(-2M\Delta^2)$ without the η factor. The gap between $1/\eta$ and $1/2$ can be arbitrarily large as $\eta \rightarrow 0$. This does not contradict the rate optimality claim (which focuses on the exponent in $M\Delta^2$), but it means the lower bound is not tight in η , and the claim of “matching” in the comparison table (Section 5.3) is overstated for the pre-factor. The author correctly notes this as a limitation.

5. Summary: What Would Need to Be Fixed

Required for Publication

1. **Lemma 7 must be replaced or rigorously proved.** The current statement is false. The author should either (a) derive the correct bound from Slud’s Gaussian comparison, which would give exponent $M\Delta^2/(2\mu(1-\mu))$ instead of $2M\Delta^2$, and then argue that the minimax lower bound still holds with exponent $2M\Delta^2$ (perhaps via a different construction or by optimizing over μ); or (b) prove a novel inequality that avoids the Gaussian intermediate. The numerical counterexamples must be addressed.
2. **The $K=2$ to $K>2$ reduction must be re-argued.** The current χ^2 -based argument uses the direction of the bound incorrectly. The author should simply note that under $C_{\text{bal}} = 1$, P_1 is a product distribution (all mixture components are identical), and the per-component TV is $|1 - \mu K/(K-1)|$, which is monotone in K and minimized at $K = 2$. No χ^2 bounding is needed.
3. **The F1 derivation (Section 5.1) must be cleaned up.** The algebraic errors in the denominator expression should be corrected. The validity condition $\text{FP} \geq \text{FN}$ (equivalently $(1-\eta)\text{FPR} \geq \eta(1-\text{TPR})$) should be stated and verified for the specific construction. If the bound is not universal, this should be noted explicitly.

Cosmetic but Important

4. **The ceiling issue** in the Slud application should be handled properly (for odd M , either use $k = (M-1)/2$ and a different detection

threshold, or use the correct $k = \lceil M/2 \rceil$ and adjust the exponent).

5. **The η gap** should be stated more carefully. The claim that the exponent “matches” is correct, but the pre-factor comparison table should clearly state that the η dependence is unresolved.

6. Bottom Line

The proof attempts to show that $\exp(-2M\Delta^2)$ is the minimax-optimal rate. The construction and the Le Cam framework are appropriate. The mixture-product TV bound, the χ^2 tensorization, and the Cramer-Chernoff refinement are technically correct.

However, the proof collapses at Lemma 7. The claimed inequality is not a valid consequence of Slud (1977) and is numerically false for parameters required by the proof. Without a valid Binomial tail lower bound, the entire lower bound on the testing error is unsupported. The remaining issues ($K > 2$ reduction, F1 derivation) are fixable with rewriting, but Lemma 7 is fundamental and cannot be patched with cosmetic changes.

Recommendation: Major revision required. The author should either (a) derive the correct lower bound using the Gaussian comparison form of Slud’s inequality, which will yield a μ -dependent exponent, and then show that the minimax rate $2M\Delta^2$ still follows (perhaps by a different argument); or (b) use an alternative technique (e.g., Le Cam combined with a direct TV calculation for products) that avoids the Binomial tail bound entirely.